Grade 10 – Term 1
Comparing Decision Criteria: C8 Practice

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Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

1 Introduction

All problems use **profit payoffs**: larger values are better, and negative values represent losses. Alternatives A_i occupy rows; mutually exclusive states of nature S_j occupy columns. The payoff $P_{ay}(A_i, S_j)$ is the profit if A_i is chosen and S_j occurs. A decision is made before the state is known.

1.1 Four ways to compare the same alternatives

Maximax selects the alternative with the highest row maximum. It asks, "Which alternative offers the greatest possible profit?" **Maximin** selects the alternative with the highest row minimum. It asks, "Which alternative has the best worst-case profit?" Neither needs probabilities, even when probabilities are available.

Minimax Regret first compares alternatives within each state:

$$b(S_j) = \max_i P_{ay}(A_i, S_j), \qquad r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

Regret is the profit forgone compared with the best alternative for the state that actually occurs; it is not necessarily a cash loss. Select the alternative with the *smallest maximum regret*. This criterion needs the full payoff table, but no probabilities.

EMV/Bayes selects the highest Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P_{ay}(A_i, S_j), \qquad P(S_j) \geq 0, \quad \sum_{j=1}^{n} P(S_j) = 1.$$

It requires a probability for every state as well as the payoffs. In these problems the state probabilities are the same for every alternative. Maximizing expected money expresses a risk-neutral orientation: it does not directly penalize uncertainty or protect against a large loss. An EMV is a weighted average, not a guaranteed profit or necessarily an attainable single outcome.

Criterion	Information and selection	Risk orientation	Main limitation
Maximax	Payoffs; highest row maximum	Optimistic; pursues upside	Ignores other outcomes and their likelihoods.
Maximin	Payoffs; highest row minimum	Conservative; protects the worst payoff	Ignores upside and how unlikely the worst state may be.
Minimax Regret	Full payoff table; smallest row maximum regret	Cautious about missed opportunities	Ignores likelihoods; depends on the alternatives included.
EMV/Bayes	Payoffs and valid probabilities; highest EMV	Risk-neutral about money	Sensitive to probability estimates; may recommend an unaffordable downside.

Criteria can disagree because they compare different summaries of the *same* information. Agreement does not mean that their reasoning is identical. Report all tied alternatives; do not invent a tie-breaking rule. Without probability information, EMV cannot be calculated unless an additional probability assumption is explicitly justified. Equal probabilities are not implied by uncertainty.

How to complete the numerical problems. The summary tables are already completed; do not recalculate row maxima, row minima, regrets, or EMVs. Identify the recommended alternative(s) by comparing the appropriate summary column. In the answer table, record the winning summary value and name the quantity it represents (for example, "Row minimum: 35"). Then justify and compare the recommendations using the supplied summaries and the original payoffs.



2 Comparing Decision Criteria

Problems:

2.1. Worked comparison: delivery expansion

2.2. Festival catering

2.3. Retail plans: agreement and ties

2.4. Bakery production

2.1 Delivery Expansion

A delivery firm chooses its expansion plan. Quarterly profits are in thousand US dollars. Forecast probabilities are shown in the table.

Alternative	Strong (S_1)	Stable (S_2)	Weak (S_3)
$P(S_j)$	0.20	0.60	0.20
Large fleet (A_1)	100	20	-20
Flexible fleet (A_2)	75	50	10
Current fleet (A_3)	40	40	35
Contract partners (A_4)	60	65	20

Evaluating these payoffs and forecast probabilities under standard decision criteria yields the following summary statistics:

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	100	-20	55	28
A_2	75	10	25	47
A_3	40	35	60	39
A_4	65	20	40	55

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Solution

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 100
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 25
EMV/Bayes	A_4	EMV: 55

Maximax selects A_1 (highest payoff: 100), despite a minimum of -20. Maximin selects A_3 (highest minimum: 35), sacrificing potential gains. Minimax Regret selects A_2 (lowest maximum regret: 25), but its minimum payoff is only 10. EMV/Bayes selects A_4 (highest EMV: 55), although its actual payoffs are 60, 65, and 20.

Maximax is optimistic; Maximin is conservative; Minimax Regret minimizes missed opportunities; EMV/Bayes maximizes expected profit using state probabilities. Only EMV uses probabilities, while regret depends on competing

alternatives. None guarantees a required profit. If avoiding losses is mandatory, exclude A_1 before applying any criterion.



2.2 Festival Catering

Use the completed summary below to identify all four recommendations in the blank answer table. A caterer chooses a large or small contract before knowing attendance. Profits are in thousand US dollars.

Alternative	High (S_1)	Low (S_2)
$P(S_j)$	0.30	0.70
Large contract (A_1)	80	10
Small contract (A_2)	45	35

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	80	10	25	31
A_2	45	35	35	38

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

Solution

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 80
Maximin	A_2	Row minimum: 35
Minimax Regret	A_1	Maximum regret: 25
EMV/Bayes	A_2	EMV: 38

Maximax selects the highest row maximum (80); Maximin selects the highest row minimum (35); Minimax Regret selects the smallest maximum regret (25); EMV/Bayes selects the highest EMV (38). There are no ties.



2.3 Retail Plans: Agreement and Ties

Use the completed summary below to identify all four recommendations in the blank answer table. A retailer compares three plans. Profits are in thousand CNY.

Alternative	Strong (S_1)	Normal (S_2)	Weak (S_3)
$P(S_j)$	0.25	0.50	0.25
Plan A_1	60	45	30
Plan A_2	50	40	20
Plan A_3	60	35	30

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	60	30	0	45
A_2	50	20	10	37.5
A_3	60	30	10	40

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

Solution

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1 and A_3	Row maximum: 60
Maximin	A_1 and A_3	Row minimum: 30
Minimax Regret	A_1	Maximum regret: 0
EMV/Bayes	A_1	EMV: 45

Maximax selects the highest row maximum (60), shared by A_1 and A_3. Maximin selects the highest row minimum (30), also shared by A_1 and A_3. Minimax Regret selects A_1 with the smallest maximum regret (0), and EMV/Bayes selects A_1 with the highest EMV (45). Both tied alternatives must be reported for Maximax and Maximin.



2.4 Bakery Production

Use the completed summary below to identify all four recommendations in the blank answer table. A bakery selects a daily production plan before demand is known. Profits are in hundred US dollars.

Alternative	High (S_1)	Normal (S_2)	Low (S_3)
$P(S_j)$	0.30	0.50	0.20
Large batch (A_1)	90	30	-10
Flexible batch (A_2)	65	50	20
Small batch (A_3)	40	40	35

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	90	-10	45	40
A_2	65	20	25	48.5
A_3	40	35	50	39

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks

C8 Which alternatives does each criterion recommend? Complete the answer table.

Solution

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 90
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 25
EMV/Bayes	A_2	EMV: 48.5

Maximax selects the highest row maximum (90), despite a possible loss of 10. Maximin selects the highest row minimum (35). Minimax Regret and EMV/Bayes agree on A_2, using the smallest maximum regret (25) and highest EMV (48.5), respectively. There are no ties; only EMV uses the demand probabilities.

3 Contextual Practice

Problems:

3.1. Urban farm crop mix

3.2. Business reserve investment

3.3. Coastal ferry capacity

3.4. Printing workshop equipment

Use the supplied summaries to complete each answer table, including ties. Justify your answers with numerical evidence; do not recalculate the summaries.

3.1 Urban Farm Crop Mix

Problem Description

An urban farm chooses a crop mix for the next season. Profits are in thousand US dollars; the estimates come from a market forecast.

Alternative	Strong (S_1)	Stable (S_2)	Weak (S_3)
$P(S_j)$	0.50	0.30	0.20
Specialty crops (A_1)	80	40	-10
Balanced crops (A_2)	60	55	20
Staple crops (A_3)	40	45	35

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	80	-10	45	50
A_2	60	20	20	50.5
A_3	45	35	40	40.5

Criterion	Recommended Alternative(s)	Deciding Value
Maximax		
Maximin		
Minimax Regret		
EMV/Bayes		

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

C8 The farm needs at least 30 in profit in every state to meet its loan payments. Specialty crops also require refrigerated storage that is unavailable this season; balanced and staple crops use existing facilities. Which mixes satisfy these restrictions? Recommend a feasible mix and explain whether securing refrigeration alone would make either rejected mix acceptable.

Solution

1. The winning values in each summary column give:

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 80
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 20
EMV/Bayes	A_2	EMV: 50.5

   There are no ties. Maximax favors A_1's upside of 80 despite its minimum of -10; Maximin favors A_3's minimum of 35. Minimax Regret and EMV/Bayes both select A_2, but use different values: maximum regret 20 and EMV 50.5.

2. Only A_3 satisfies both restrictions: it uses existing facilities and its minimum profit of 35 covers the required 30. Reject A_1 because it lacks refrigeration and can lose 10; reject A_2 because its minimum of 20 cannot cover the loan payments. Recommend staple crops despite their lower EMV of 40.5 versus 50.5 for A_2. Refrigeration alone would not resolve either mix's profit shortfall.



3.2 Business Reserve Investment

Problem Description

A business invests part of its reserve for one year. The table gives *net gains*, in thousand US dollars, rather than final balances. All listed states are possible.

Alternative	Growth (S_1)	Stable (S_2)	Recession (S_3)
$P(S_j)$	0.60	0.25	0.15
Growth fund (A_1)	90	25	-30
Balanced fund (A_2)	60	40	5
Fixed deposit (A_3)	25	25	25

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	90	-30	55	55.75
A_2	60	5	30	46.75
A_3	25	25	65	25

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks

C8 Which alternatives does each criterion recommend? Complete the answer table.

C8 The owner requires nonnegative gains in every state and access to the invested reserve for emergency payroll. Both funds permit withdrawals within two business days, but the fixed deposit locks the money for the full year. Which investments meet both requirements? Recommend a feasible investment and explain when the fixed deposit could become acceptable.

Solution

1. The winning values in each summary column give:

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 90
Maximin	A_3	Row minimum: 25
Minimax Regret	A_2	Maximum regret: 30
EMV/Bayes	A_1	EMV: 55.75

   There are no ties. Maximax and EMV/Bayes both select A_1: one favors its maximum of 90, the other its EMV of 55.75. Their optimistic and risk-neutral priorities remain distinct; Maximin protects 25, while Minimax Regret limits missed opportunities to 30.

2. Only A_2 meets both requirements: its minimum gain is 5 and withdrawals take two business days. Reject A_1 because it can lose 30, and A_3 because its lock prevents emergency access despite its guaranteed gain of 25. Recommend the balanced fund, provided payroll can wait two business days. The fixed deposit could become acceptable if a separate liquid reserve fully covers emergency payroll; its lower EMV buys stronger protection of the annual gain.

3.3 Coastal Ferry Capacity

Problem Description

A ferry operator selects a seasonal capacity plan. Profits are in thousand US dollars. No reliable probabilities are available.

Alternative	High (S_1)	Normal (S_2)	Low (S_3)
Add two ferries (A_1)	100	35	-25
Add one ferry (A_2)	75	55	10
Lease peak capacity (A_3)	55	48	30
Keep current fleet (A_4)	30	30	30

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	100	-25	55	—
A_2	75	10	25	—
A_3	55	30	45	—
A_4	30	30	70	—

EMV cannot be determined because state probabilities are unavailable. Mark the EMV/Bayes recommendation and deciding value as unavailable.

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

C8 Only one additional qualified crew is available, so adding two ferries is impossible. Leased peak capacity requires a berth permit that will not arrive before the season; adding one ferry and keeping the current fleet need no new permit. Which plans are feasible? Recommend one, weighing expansion against reliable service, and explain what must change before leasing can be considered.

Solution

1. The winning values in each summary column give:

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 100
Maximin	A_3 and A_4	Row minimum: 30
Minimax Regret	A_2	Maximum regret: 25
EMV/Bayes	Unavailable	Unavailable

Maximax favors the upside of 100, Maximin protects the minimum of 30, and Minimax Regret limits missed opportunities to 25. Report both A_3 and A_4 for Maximin: their minima match despite different upsides. Without

state probabilities, the plans' EMVs cannot be compared; A_4's constant profit of 30 alone does not establish a Bayes choice.

2. Only A_2 and A_4 are feasible: A_1 lacks a second additional crew and A_3 lacks its berth permit. One justified choice is A_4 to prioritize reliable service with familiar operations and a minimum profit of 30. Choose A_2 instead if the available crew can support expansion reliably and the operator accepts a minimum of 10 for greater upside. Leasing can be reconsidered once the permit is issued in time for service; its payoff alone cannot make it operationally feasible.



3.4 Printing Workshop Equipment

Problem Description

A printing workshop selects an equipment plan for the next year. Net profits are in thousand US dollars and already include all equipment costs.

Alternative	High (S_1)	Normal (S_2)	Low (S_3)
$P(S_j)$	0.40	0.40	0.20
Automated line (A_1)	100	50	0
Standard upgrade (A_2)	75	55	25
Maintain equipment (A_3)	50	45	40

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	100	0	40	60
A_2	75	25	25	57
A_3	50	40	50	46

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

Questions/Tasks



C8 Which alternatives does each criterion recommend? Complete the answer table.

C8 The workshop has 60 available for upfront spending and cannot borrow. The automated line requires 80, the standard upgrade 50, and maintaining equipment 30, all in thousand US dollars. Which plans are feasible? Recommend a feasible plan, compare its EMV and minimum profit with the other feasible plan, and justify your priority. Do not deduct upfront spending again from the listed profits.

Solution

1. The winning values in each summary column give:

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 100
Maximin	A_3	Row minimum: 40
Minimax Regret	A_2	Maximum regret: 25
EMV/Bayes	A_1	EMV: 60

   There are no ties. Maximax and EMV/Bayes select A_1 for its maximum of 100 and EMV of 60. Maximin protects a minimum of 40 with A_3; Minimax Regret selects A_2 to limit missed opportunities to 25.
2. Only A_2 and A_3 are feasible: their upfront costs of 50 and 30 fit the budget of 60; A_1 exceeds it by 20. One justified choice is A_2 for expected profit: its EMV of 57 exceeds A_3's 46 by 11, but its minimum of 25 is 15 lower than A_3's 40. Choose A_3 instead if worst-case protection is the priority. The highest overall EMV of 60 cannot make A_1 affordable.



4 Mixed Practice

Problems:

4.1. Choosing a criterion

4.2. Evaluating decision claims

4.3. Decision committee report

4.1 Choosing a Criterion

Which criterion fits each *stated priority*? Give its required information and one limitation, and briefly contrast another feasible criterion.

1. A start-up knows its payoffs but lacks reliable probabilities. Which criterion suits a founder seeking maximum upside and accepting the downside?
2. A family business has reliable probabilities. Which criterion suits its priority of the highest worst-case profit?
3. A delivery manager lacks reliable probabilities. Which criterion minimizes the greatest profit forgone relative to the best plan in each state?
4. A retailer trusts its probabilities and makes repeated seasonal decisions. Which criterion suits its priority of average monetary profit?
5. A museum knows its payoffs but lacks attendance probabilities. What information is missing for an EMV choice, and why would equal probabilities be an assumption?

Solution

1. Maximax fits the optimistic priority and needs only payoffs, but ignores downside outcomes. Maximin is also feasible and instead protects the worst payoff.

2. Maximin fits the conservative priority and needs only payoffs, but ignores upside and likelihoods. EMV/Bayes is feasible but prioritizes expected profit.
3. Minimax Regret needs the full payoff table, without probabilities, but may sacrifice worst-case protection. Maximin is feasible and instead protects the minimum profit.
4. EMV/Bayes fits the risk-neutral priority and needs payoffs and reliable probabilities, but cannot guarantee a season's profit. Maximin is feasible and instead protects the worst outcome.
5. A nonnegative probability for each attendance state, summing to one, is missing. Equal probabilities would assume equally likely states without evidence. The three payoff criteria remain feasible but answer different priorities.



4.2 Evaluating Decision Claims

Is each claim valid? Correct it if needed, using a numerical example where possible.

1. "Maximin and Minimax Regret always agree because both are cautious."
2. "An EMV of 55 guarantees a seasonal profit of 55."
3. "Regret of 25 means an actual monetary loss of 25."
4. "Agreement with Maximax makes Bayes optimistic."
5. "Valid probabilities make EMV best for every decision-maker."

Solution

1. Invalid. In delivery expansion, Maximin selects A_3 (minimum 35); Minimax Regret selects A_2 (maximum regret 25, minimum 10). Worst-case protection and limited missed opportunities are different priorities.
2. Invalid. Delivery plan A_4 has EMV 55 but can earn only 60, 65, or 20. Expected profit is not guaranteed profit.
3. Invalid. With low festival attendance, A_1 earns 10 but has regret $35 - 10 = 25$. Regret measures forgone profit, not an actual loss.
4. Invalid. Both choose investment A_1, but Maximax uses its maximum of 90 and Bayes its EMV of 55.75. Bayes remains risk-neutral.
5. Invalid. The farm's EMV choice A_2 has minimum 20 and fails the required 30. Only A_3, with minimum 35 and EMV 40.5, meets the obligation despite valid probabilities.



4.3 Decision Committee Report

Return to the delivery expansion problem in Section 2.1. Four committee members prioritize, respectively, the largest possible profit, the best worst-case profit, the smallest maximum missed opportunity, and the highest expected profit.

1. Which criterion and plan fit each member's priority? Give the deciding value and information used or ignored.
2. Which plan would you recommend? Compare two plans numerically, state your priority and one limitation, and explain a reasonable disagreement.
3. If the forecast is withdrawn, which summaries remain usable and which recommendation loses support? Does one remaining criterion become universally best?

Solution

1. In the stated order: Maximax chooses A_1 (row maximum 100), ignoring downside and probabilities; Maximin chooses A_3 (row minimum 35), ignoring upside and probabilities. Minimax Regret chooses A_2 (maximum regret 25), comparing all payoffs but ignoring likelihoods; EMV/Bayes chooses A_4 (EMV 55), using payoffs and probabilities without protecting the worst outcome.
2. One valid report is: "We prioritize expected profit and recommend A_4: its EMV is 55 versus 47 for A_2. This depends on reliable probabilities and does not guarantee 55. A member prioritizing missed opportunities may prefer A_2, with maximum regret 25 versus 40 for A_4." Other choices are valid with a stated priority, numerical support, and a limitation.
3. Row maxima, row minima, and maximum regrets remain usable, retaining choices A_1, A_3, and A_2, respectively. The EMVs and Bayes choice A_4 lose their forecast basis. No remaining criterion becomes universally best: each serves a different priority.

Self-check. State the information, priority, recommendations (including ties), deciding values, and limitations.

